

MACD

Indicators & Analysis / Momentum & Oscillators | beginner | 5 min read | DeepCharts Help Center | July 16, 2026

MACD (Moving Average Convergence Divergence) turns the relationship between two exponential moving averages into a compact momentum display: a MACD line, a signal line and a histogram. Developed by Gerald Appel in the 1970s, it remains one of the most widely used indicators in technical analysis.

This reference covers adding MACD to a DeepCharts chart, reading its three components, and every option in its settings dialog.

What it is

MACD answers one question: is short-term momentum strengthening or weakening relative to the longer-term trend? It compares a fast exponential moving average (12 periods by default) with a slow one (26 periods by default). The MACD line tracks the distance between the two averages, the signal line is a smoothed companion to the MACD line, and the histogram shows the gap between the two — a quick visual for momentum building or fading.

Because it is built entirely from moving averages, MACD is a lagging indicator: it confirms momentum shifts rather than predicting them.

When to use it

- You want one panel that shows both trend direction and momentum strength.
- You need confirmation before acting on a price level or pattern — a MACD cross in the same direction adds weight.
- You are watching for divergence: price making a new extreme while momentum does not.
- You want an objective definition of "momentum turned" (a line crossing) instead of eyeballing candles.

Quick start

1. Open a chart and press **Ctrl I** — or click the bar-chart icon in the chart toolbar, then **Indicators** — to open the **Indicator List**.
2. Type "MACD" in the search box and click **+** next to the indicator. MACD renders in its own chart area below the price bars (see [How to Change the Layout of Indicators](#) to move or resize it).
3. Click the gear icon on the indicator's row to open its settings.
4. For a first configuration, keep the defaults: **Fast EMA Period 12**, **Slow EMA Period 26** and **Input Data** set to Close. You can try those choices in the live MACD below — drag the fast, slow and signal sliders and watch how the histogram and crossover count respond.

[WIDGET: macd-lab]

How to read it

- **Zero line.** MACD above zero means the fast average is above the slow one — momentum is on the bullish side. Below zero, the bearish side.
- **Crossovers.** The MACD line crossing above the signal line marks momentum turning up; crossing below marks it turning down. Crossovers that happen far from the zero line carry more weight than ones right at it.
- **Histogram.** Each bar is the distance between the MACD and signal lines. Shrinking bars mean the move is losing momentum, and often precede a crossover by a few bars.
- **Divergence.** Price prints a new high while MACD prints a lower high (or the mirror image at lows) — the move is running on less momentum and is more vulnerable to a reversal.

[SCREENSHOT PLACEHOLDER] Close-up of the MACD chart area showing the histogram shrinking into a signal-line crossover, with the crossover bar highlighted `macd-histogram-crossover.png`

Settings reference

Open the dialog from the gear icon on the indicator's row in the **Indicators** panel.

Parameters

Setting	What it does
Fast EMA Period	Length of the fast moving average. Default: 12. Lower it for earlier but noisier signals.
Slow EMA Period	Length of the slow moving average. Default: 26. Raise it to focus on longer swings.
Input Data	The data series fed into the calculation — Close, Open, High, Low or Volume. Close is the standard choice; see Different Types of Input Data for Indicators for when the others make sense.
Smooth	Smoothing applied to the MACD line. Keep the default unless the line is too jagged to read on fast timeframes.

[CONFIRM: whether the signal-line period is configurable in this dialog and what its default value is]

Subgraph

Setting	What it does
Histogram color	Color of the histogram bars.
Signal color	Color of the signal line.
MACD color	Color of the MACD line.
Secondary color	Optional additional color for the indicator display.
Subgraph style	How the series are drawn in the chart area.
Line style	Line rendering style for the MACD and signal lines.
Line width	Thickness of the lines.
Use secondary axis	Plots the indicator against a secondary value axis.

[SCREENSHOT PLACEHOLDER] The MACD settings dialog open on the Parameters section, showing Fast EMA Period 12, Slow EMA Period 26, the Input Data dropdown and the Smooth option `macd-settings-parameters.png`

Tips and common mistakes

- **MACD lags by design.** It is built from moving averages, so use it to confirm a move you already have a reason to expect, not to predict one.
- **Chop kills crossovers.** In a sideways market MACD hugs the zero line and whipsaws. Filter by taking only crossovers on the trend side of zero, or pair it with a Moving Average trend filter.
- **A histogram flip is not a crossover.** The histogram changes direction before the lines actually cross — treat it as an early warning, not a completed signal.
- **Absolute values are not comparable across symbols.** MACD is measured in price units, so a reading of 5 on one instrument tells you nothing about a reading of 5 on another. Read it within one chart only.

Related articles

- Relative Strength Index (RSI)

- Stochastic Oscillator
- Momentum
- Moving Average
- Different Types of Input Data for Indicators
- How to Change the Layout of Indicators